SUMMARY · MK 6 BUSINESS ANALYTICS · WEEK 3

Data-Driven Decision Making, Descriptive Analytics & Storytelling

Decision types and stages, intuition vs data-driven decisions, the data–intuition partnership, DIKW, analytics levels (Zara), descriptive statistics, visualisation and chart choice, data storytelling and decluttering.

Material

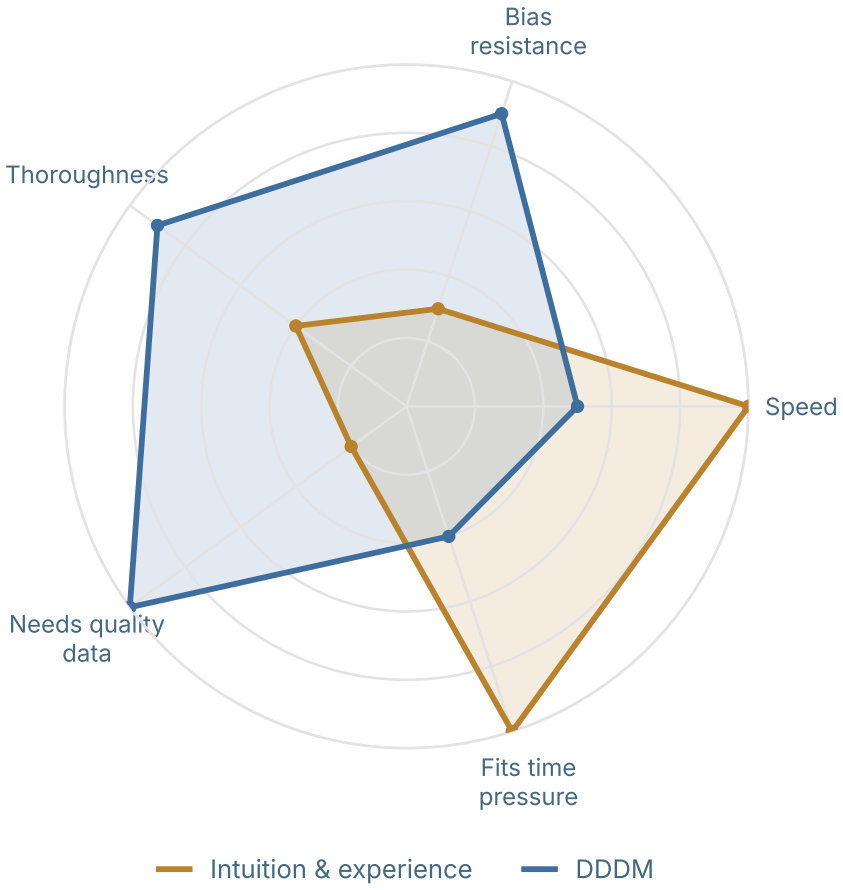
1 Decisions and DDDM

Strategic, tactical, operational decisions; five stages. Traditional (intuition/experience) vs DDDM (validate before acting: surveys, user tests, test markets). Data and intuition form a feedback loop.

2 Intuition and data as partners Deep dive

Intuition asks → Data tests → Results refine intuition → Intuition flags bad data

Traditional vs data-driven (conceptual)

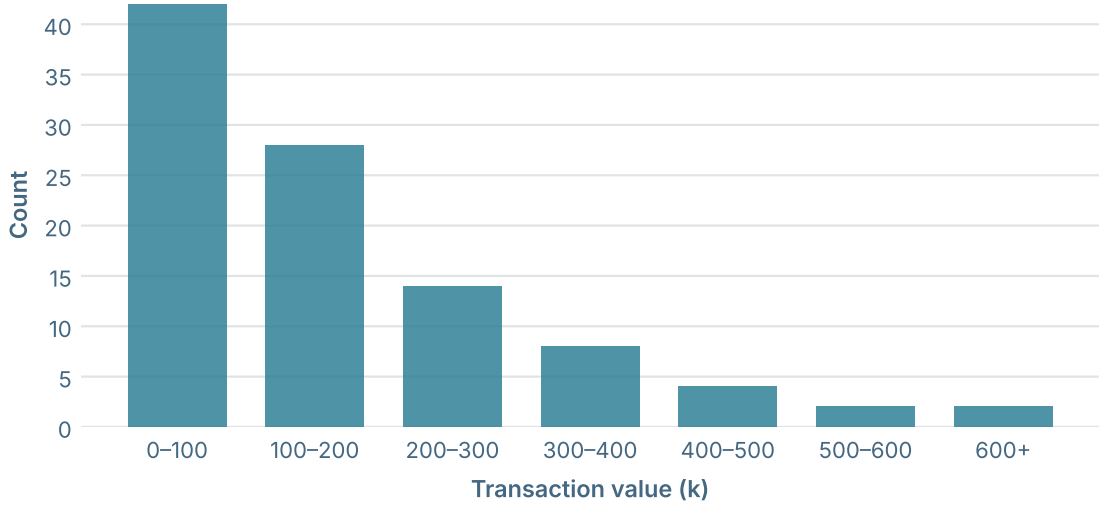


3 Descriptive analytics and statistics

Descriptive summarises what happened; diagnostic asks why. Centre, spread, shape, relationships. Zara uses all four levels for fast fashion supply.

4 Descriptive statistics that belong together Deep dive

A right-skewed distribution (illustrative)



Mode 0–100, median ~130, mean ~180.

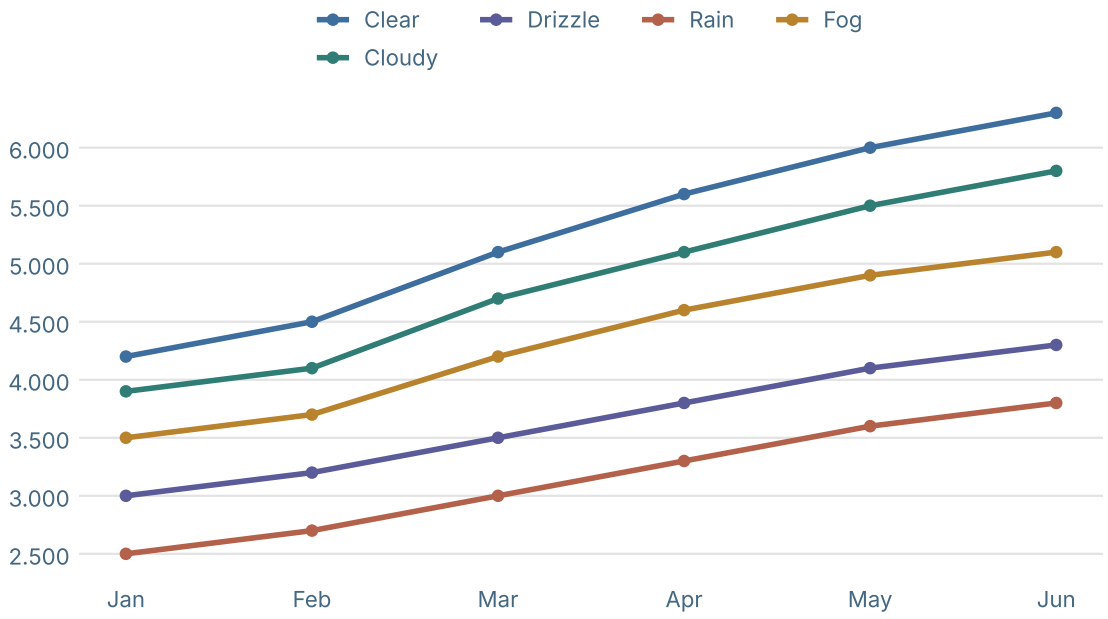
Right-skewed: mode < median < mean.

5 Visualisation and storytelling

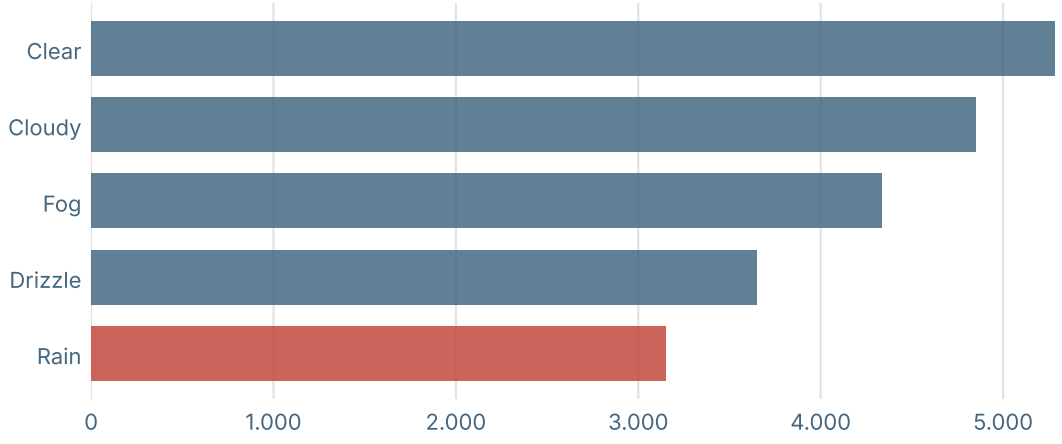
Choose charts by message (bar, line, stacked, histogram, scatter, map); avoid 3D, truncated bars, rainbow colours. Storytelling: context → finding → action; stories are ~22× more memorable. Declutter and use conclusion titles.

6 Before and after: from cluttered to a story Deep dive

Before: 'Bike rentals by weather'



After: 'Rainy days cut rentals ~40%'

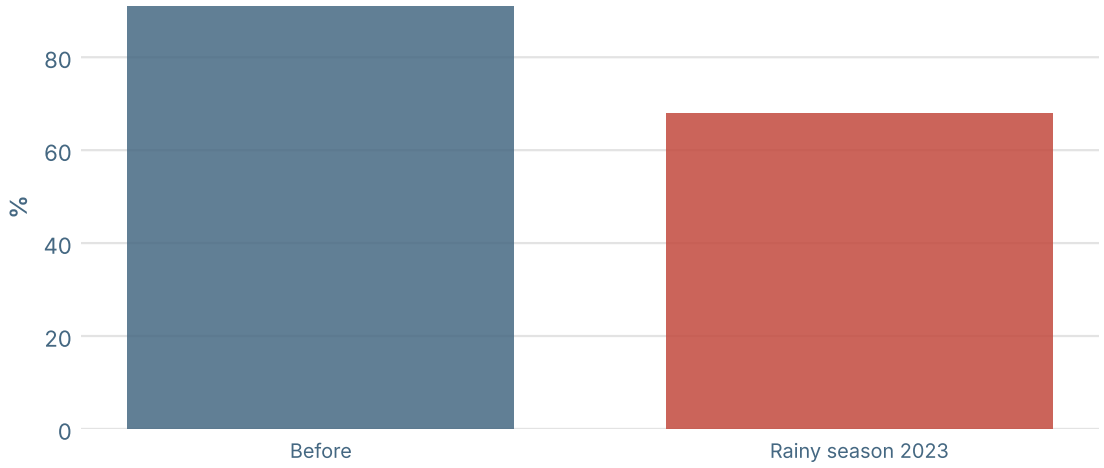


Conclusion title, one highlighted category (illustrative data).

7 For BetonPrima: two charts that convince the board

Deep dive

West Java on-time delivery fell 23 points in the rainy season



Order-to-site rose to 5.4 days (target 3.0).

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Context → finding → action
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Key takeaways

1. Decision types.
2. DDDM validates first.
3. Data and intuition partner.
4. Describe centre, spread, shape.
5. Right chart, no clutter.
6. Context → finding → action.

Case studies

Case & worked answers

Bike sharing: from cluttered chart to story

Redesign the bike-sharing chart: one message, conclusion title, highlight rain, end with an action. Apply to BetonPrima: on-time decline line with rainy-season

annotation plus churn-reason bars.

Practice questions

Try answering first. The answer key is below each question.

1. Best chart for a time trend...

A. Pie
B. Line
C. Scatter
D. Map

Answer: B. Line.

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2. A good storytelling title...

A. 'Rental chart'
B. 'Rain cuts rentals 40%'
C. '2024 data'
D. None

Answer: B. Title = conclusion.

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3. In right-skewed data, usually...

A. Mean < median < mode
B. Mode < median < mean
C. All equal
D. Median < mode < mean

Answer: B. The tail pulls the mean.

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4. Zara's fast allocation is...

A. Descriptive
B. Diagnostic
C. Predictive
D. Prescriptive

Answer: D. Decides action.

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5. DDDM needs all EXCEPT...

A. Clear goals
B. Quality data
C. Interpretation skills
D. Removing intuition entirely

Answer: D. They're partners.

Sources

- Week 3 MK6 Material — Descriptive and Diagnostic Analysis
- Knaflic, Storytelling with Data